

2025OS - VMware Workstation

文件(F) 编辑(E) 查看(V) 虚拟机(M) 选项卡(T) 帮助(H)

库

我的计算机

- CS323-Compile
- aaa
- Ubuntu 64 位
- 2025OS

2025OS

主页 2025OS

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Apr 3 17:45

trap.c - assignment1-Preemption(2)(1) - Visual Studio Code

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EXPLORER

- ASSIGNMENT1-PREEMPTION(2)(1)
 - os
 - kalloch
 - kernel.ld
 - lock.c
 - lock.h
 - log.h
 - main.c
 - memlayout.h
 - nommu_init.c
 - plix.c
 - plix.h
 - printf.c
 - printf.h
 - proc.c
 - proc.h
 - queue.c
 - queue.h
 - riscv.h
 - sbi.c
 - sbi.h
 - sched.c
 - shm.c
 - string.c
 - string.h
 - switch.S
 - syscall_ids.h
 - timer.c
 - timer.h
 - trap.c
 - trap.h

C trap.c

- os > C trap.c > kernel_trap(ktrapframe *)
- 23 void kernel_trap(struct ktrapframe *ktf) {
- 26 if ((r_sstatus() & SSTATUS_SPP) == 0) {
- 27 errorf("kerneltrap: not from supervisor mode");
- 28 goto kernel_panic;
- 29 }
- 30 }
- 31 mycpu()->inkernel_trap++;
- 32 }
- 33 uint64 cause = r_scause();
- 34 uint64 exception_code = cause & SCAUSE_EXCEPTION_CODE_MASK;
- 35 if (cause & SCAUSE_INTERRUPT) {
- 36 // correctness checking:
- 37 if (mycpu()->inkernel_trap > 1) {
- 38 // should never have nested interrupt
- 39 print_sysregs(true);
- 40 print_ktrapframe(ktf);
- 41 errorf("nested kerneltrap");
- 42 goto kernel_panic;
- 43 }
- 44 if (panicked) {
- 45 panic("other CPU has panicked");
- 46 }
- 47 }

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS

[sched 0,-1] scheduler: switch to proc pid(2)

[WARN 0,2] worker: thread 2: exiting

[sched 0,2] sched: switch to scheduler pid(2)

[sched 0,-1] scheduler: switch to proc pid(3)

[ERROR 0,3] kernel_trap: kerneltrap: not from supervisor mode

[ERROR 0,3] kernel_trap: Kernel Panic

sstatus : 0x8000000200000020

- SUM:0, SPP:U, SPIE:1, SIE: 0

scause : 0x0000000000000002

- Interrupt:0, Code:2

sepc : 0x0000000000020197c

stval : 0x00000000100024f3

sip : 0x0000000000000000

- Pending: Software:0, Timer:0, External:0

Ln 56, Col 18 Spaces: 4 UTF-8 LF () C Linux

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switch.S - assignment1-Preemption(2) - Visual Studio Code

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EXPLORER

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 - string.h
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 - syscall_ids.h
 - timer.c
 - timer.h
 - trap.c
 - trap.h
- OUTLINE
- TIMELINE

os > switch.S

9 switch:

24 csrr t0, sstatus

25 sd t0, 112(a0)

26 csrr t0, sepc

27 sd t0, 120(a0)

28

29 ld ra, 0(a1)

30 ld sp, 8(a1)

31 ld s0, 16(a1)

32 ld s1, 24(a1)

33 ld s2, 32(a1)

34 ld s3, 40(a1)

35 ld s4, 48(a1)

36 ld s5, 56(a1)

37 ld s6, 64(a1)

38 ld s7, 72(a1)

39 ld s8, 80(a1)

40 ld s9, 88(a1)

41 ld s10, 96(a1)

42 ld s11, 104(a1)

43 ld t0, 112(a1)

44 csw sstatus, t0

45

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS

[INFO 2,1] init: thread 11 exited with code 31, expected 31

[INFO 2,1] init: thread 12 exited with code 32, expected 32

[INFO 2,1] init: thread 13 exited with code 33, expected 33

[INFO 2,1] init: thread 14 exited with code 34, expected 34

[INFO 2,1] init: thread 15 exited with code 35, expected 35

[INFO 2,1] init: thread 16 exited with code 36, expected 36

[INFO 2,1] init: all threads exited, count 150000

[INFO 2,1] init: init ends!

[PANIC 2,1] os/proc.c:226: init process exited

[PANIC 3,-1] os/trap.c:45: other CPU has panicked

[PANIC 1,-1] os/trap.c:45: other CPU has panicked

[PANIC 0,-1] os/trap.c:45: other CPU has panicked

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